FS26 Resources List

General Resources

Important Organizations with Resources

The Abdul Latif Jameel Poverty Action Lab (J-PAL)

The Abdul Latif Jameel Poverty Action Lab is premier global research center specializing in randomized evaluations in development economics and policy. They provide multiple resources on RCTs with specific resources available on energy and climate change. Their catalogue includes events and coding resources to help with each step of the process of designing an experiment and evaluating the data.
Some recommended resources include:

- Research Resource: https://www.povertyactionlab.org/research-resources
- The GitHub has coding resources for randomized evaluations: https://www.povertyactionlab.org/resource/coding-resources-randomized-evaluations
- A guideline for Sample Size and Power Calculations to help in experimental planning https://github.com/J-PAL/Sample_Size_and_Power
- A general data analysis. https://www.povertyactionlab.org/resource/data-analysis

Similar to J-PAL, but focused on poverty, is the Innovations for Poverty Action group. J-PAL appears to have more resources, but this group could have some events of interest for you: https://www.poverty-action.org

Innovation Growth Lab (IGL)

The Innovation Growth Lab represents a global collaboration that develops and tests different approaches to support innovation, entrepreneurship, and growth. Their focus is also experimental methods in policy evaluation. They also have a collection of resources for analysis and experimental design, and looking through their catalogue of previous projects can help generate new ideas for an experiment or a method to use.

Resources: https://www.innovationgrowthlab.org/resources
Recommendations:

- Designing Field Experiments – An IGL Masterclass Series
- Quantitative analysis guide
- Trial protocol and statistical analysis plan templates

- Example Study - Training Programme: Driving policy impact through experimentation
  A video outlining the main goals of the IGL : Experimentation for innovation.
  https://www.youtube.com/watch?v=8gpF1lpZ7uw

Topic References

Pre-Analysis Plans

J-PAL provides an excellent description of how pre-registrations currently work, some alternatives being developed, and how to write one for your experiment. They also provide a series of examples to reference: https://www.povertyactionlab.org/resource/pre-analysis-plans

This one is a particularly good example for examining how to develop your model specification sections. Its pre-registration can be found here, which contains its pre-analysis plan as well. Study actually aims to review the

Pre-registrations

Here are three AEA pre-registrations that provide a better structure than the previous, energy-related examples can be found here:

1. "Making inequality unacceptable: an experiment on fiction reading and prosociality" provides a short but well structured pre-registration.
2. Enhancing Equity and Sustainability in Utility Assistance Programs provides a slightly longer, but still well structured. The subject is somewhat similar to the weatherization paper we discussed in class on 12.11, and can be connected to the overall energy subject of the course.
3. Sugar Policy and Sustainability – Numeric Calorie Labels, Symbolic Calorie Labels, and No-Sugar Subsidy" is a longer-form structure, for an alternativeto the others we have provided. It examines nudge interventions.

A discussion of the issue of p-hacking in economics, one of the motivations for pre-registrations, can be found here.

Literature Search Tools

Recently, we stumbled across this tool for searching the economics literature for empirical methods. This can also be used to search for keywords of interest that are not just empirical methods. The tool searches the entire text of the papers in its database and provides a list of those that match your criteria, as well as a plot of the proportion of papers in the database that use that term on an annual basis. You can explore it here: paulgp.com/econlit-pipeline/index.html.

Another tool which provides nice connections of papers across all fields is Connected Papers. Here, you can visually search the literature by displaying citation as a graphical network.

Literature Review

The article Ten Simple Rules for Writing a Literature Review provides a clear outline in an easy-to-read setup for people learning about how to ensure a literature review remains structured whilst covering an extensive number of papers.

How to Do a Systematic Review provides distinctions between different types of literature reviews. It also introduces meta-analysis approaches to provide a robust outline of the quantitative and qualitative approaches to a literature review.

Doing Meta-Analysis with R is a detailed guide through meta-analysis, which focuses on the quantitative tools and concepts necessary to review the stated effect sized from different papers. This focuses on "non-expert" backgrounds in statistical and programming background, making it a good fit for an introductory textbook with coding examples in R in every section.

A living literature review is a newer model of literature review which avoids the issue of doubling work from different teams over a short period of time. Examples can be found here. There are comparable examples for living meta-analyses, and these came into prominence during COVID-19 and the related research during that time.

Experimental Design

Evaluating What Works is a textbook written by Dorothy Bishop which describes developing an evaluation of a policy in detail. (Typically, health policy interventions are used in this book.) Short exercises at the end of each chapter focus on the conceptual and design skills as opposed to the mathematical tools used to analyze data, which should help during the development of proposals and pre-analysis plans during this course: https://bookdown.org/dorothy_bishop/Evaluating_What_Works/

- Chapter 10 is good for power calculations, with a recommended tool to play around with to explore how standard deviation and effect size can impact power. It even allows for looking at clustered designs, and provides the opportunity to review the R code on which it is based: https://egap.shinyapps.io/Power_Calculator/

This textbook (used in my course at ETH on experimental design) provides a more theoretical text with some introductions to the mathematical components: A First Course in Design and Analysis of Experiments by Gary Oehlert

Causal Analysis

Applied Causal Analysis with R is a social-science-oriented introduction to causality that is structured and a more modular format instead of a typical textbook with longer explanations.

Epidemiology is a field where there is a lot of causal inference, and they have some excellent examples of the use of DAGs in their research. (Mostly Clinical) Epidemiology in R provides great examples of those DAGs in chapter 7 and very well documented coding examples. Other chapters are useful for learning different experimental designs when humans are the observational unit.

One of the definitive mathematical causality textbooks is Elements of Causal Inference. It provides much more depth than we use in the class, but can be quite good for developing a solid mathematical footing of the ways graph theory can be used to understand graphs. It is used in this ETH Causality lecture from 2021 during COVID, and the videos are also available from the lectures

The chapter on Directed Acyclical Graphs (DAGs) in the Causality Mixtape is a chapter I recommend reading more strongly, as this can be a good tool for conceptualizing the space you are working in for your experiment, as well as being approachable and using examples from mainstream economic literature: https://mixtape.scunning.com/

Another brand new one that is better than the Causality Mixtape and references it if you still want it is Causal Inference in R. This textbook is better structured from a mathematical framework and provides examples outside of economics, alongside additional assistance with R and tidyverse.

Coding Resources

The classic intro to R textbook you will find in most courses introducing coding for the first time (and those that make you use R). Quick and easy to reference thanks to its online design. Other good books on R also come from this author: https://r4ds.hadley.nz/

A good resource for those who know one language and want to reference the other quickly on specific topics. I have used this one myself as someone more proficient in R: For R and Python: A Tour of Data Science by Nailong Zhang (openly available as a PDF)

RCT Cookbook, providing short coding examples of how to develop a and layout a trial in both R and Stata. The GitHub of how this book was made is also available.
https://icostatistics.github.io/rct-cookbook/
https://github.com/icostatistics/rct-cookbook

Practical Issues in Research

The Catalogue of Bias discusses the biases that can arise from human behaviour. This is relevant for assessing how participants in experiments react to treatment, as well as for

reflecting upon how our own biases shape the experiments and treatments we design.

How Scientists Lie provides a catalogue of questionable research practices (QRPs), along with guidelines on how to identify and deal with them. It is critical that we examine the issues that arise in current scientific research and where they come from, whether from our own biases or the issues inherent in the current publication ecosystem.

For a more modern QRP, several sections of the MIT AI Risk Repository highlight relevant issues to consider when using chatbots or other AI tools in research workflows.

Miscellaneous

A brief introduction to the school of thinking behind behavioural economics, for those who have taken courses mostly in neoclassical economics: https://www.exploring-economics.org/en/orientation/behavioral-economics/

The Estimating the Credibility of Past Research course from the University of Bern provides a number of resources for critically reviewing individual papers. Students can reference during the process of screening papers in literature reviews

Stats with Brian is a YouTube channel which explains statistical concepts from lemmas underlying common tools to the interpretation of common tests. I highly recommend it for any confusion that can arise from denser materials